

Governance Framework for Linking Individual-level Health Data

Rule Harmonization Protocol

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¹ OS-PCORTF Project Information: <https://aspe.hhs.gov/creating-new-data-research-environment>.

Record of Changes

Version	Date	Contributors	Description of Changes
1	July 2025	G. Jenkins, E. Kraus, J. Grand, C. Rosen	Initial version.
2	September 2025	G. Jenkins, E. Kraus	Content revised and updated structure based on sponsor feedback.
3	October 2025	G. Jenkins, E. Kraus	Content revised based on sponsor feedback.
4	December 2025	G. Jenkins, E. Kraus	Updates to business process diagram and simplified language throughout document.
4.1	April 2026	G. Jenkins, E. Kraus	Updated language and formatting for consistency. Addition of cover and descriptive figures.
4.2	August 2026	E. Kraus, S. Hull	Revised rule harmonization and continuous improvement content. Added website references and section 9.

Introduction

This document is a protocol for rule harmonization, a new and structured process within the Common Governance Process to align and integrate governance rules across multiple datasets when linking individual-level health data for research. The objective of rule harmonization is to create a single consolidated rule set that provides clear, streamlined guidance on how the linked data may be created, accessed, shared, used, and disseminated. By identifying and reconciling overlapping or conflicting rules, rule harmonization advances responsible data use; legal, ethical, and organizational compliance; and risk minimization.

Background

What Is a Governance Rule?

A governance rule is a specific directive that dictates how data is accessed, shared, linked, used, and/or disseminated² within or between organizations. Governance rules are generally centered on three core concepts: the origin (where the rule originated from, such as a law), the intent (the objective the rule aims to achieve, such as ensuring the security of individual-level data), and the application (who and what action the rule applies to, such as how an investigator/requesting institution may access data).³

Governance rules are established by policies and originate from governance sources, including laws; regulations; institutional policies; project-, study-, or dataset-specific policies; standards; ethical requirements; and organizational practices and norms. Each governance source can hold one or multiple policies that assert a range of rules from broad principles for overall data management to highly specific requirements targeting a single action. Because policies are established at multiple levels (e.g., federal, state, local, and institutional), datasets and dataset holders are subject to interacting layers of governance rules. Some rules may align across datasets, while others may impose contradictory requirements that cannot be simultaneously satisfied.

What Is Rule Harmonization?

When datasets are linked, the health data about individuals from separate originating datasets is combined in a resulting linked dataset. The governance rules associated with each originating dataset remain applicable to the linked dataset, but there is no process to combine and integrate those rules into a single, cohesive set of rules governing the linked dataset. While some governance rules across datasets may align, others may overlap or conflict, creating challenges for compliance, consistency, and responsible data management. Without harmonization, the investigator/requesting institution may encounter multiple, potentially contradictory rules with no clear framework for integration.

At the same time, not every governance rule for a given dataset will apply to the creation of linked data. Many governance rules may or may not be applicable, depending on the linked

² The data lifecycle refers to the process of accessing, sharing, linking, using, and disseminating data that is fundamental to research with linked data.

³ These three concepts are further explained in [Foundation of Governance Rules](#).

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dataset context. For example, governance rules about how data is contributed to a registry are unlikely to have an application to secondary use of registry data. Thus, rule harmonization requires not only gathering governance sources and rules, but also determining applicability of a rule to a linkage request before harmonization begins.

Rule harmonization provides a structured process for systematically gathering, aligning, and reconciling governance rules across datasets to generate a consolidated rule set that guides the investigator/requesting institution in responsible, linked data creation and use. From the perspective of the dataset holder, rule harmonization is a means to collaborate and clarify the overlapping and conflicting rules to permit dataset access for linkage.

Rule harmonization seeks to model many established legal practices. For example, in corporate law, contract integration and consolidation are used to reconcile terms from multiple agreements into a single governing document. In federal systems with overlapping jurisdictions (e.g., U.S. federal and state laws), “conflict of laws” analysis is used to reconcile competing legal rules. These legal practices provide the conceptual foundation for rule harmonization.

Rule harmonization is led by the coordinating body and begins with the collection and review of governance sources. Rules are then extracted from sources and cataloged in a rule inventory for the request. Rules are evaluated for applicability and classified by domain and subdomain based on applicability to the request. Within each domain/subdomain, rules are mapped across datasets and categorized based on their interrelationships so that overlapping and conflicting rules can be harmonized into a consolidated rule set. Once approved by dataset holders and the investigator/requesting institution, the consolidated rule set can be finalized for use in the study.

The consolidated rule set serves as the foundation for creating and managing the linked dataset, offering clear guidance to investigators/requesting institutions on how the linked dataset may be accessed, shared, linked, used, and disseminated in compliance with applicable laws, ethical standards, and policies. The consolidated rule set may also serve as a basis for formal agreements that define the terms and conditions for creating and using the linked dataset, such as DUAs.

Why Is Rule Harmonization Necessary?

The governance rules that apply to each originating dataset that are to be linked are inherited by the linked dataset because governance rules follow the data.⁴ To ascertain and manage inherited rules that flow from multiple sources, investigators/requesting institutions need a process to gather and reconcile rules from originating datasets into a single, consolidated set of governance rules that harmonizes conflicting and overlapping rules.

Rule harmonization is important when linking multiple datasets because as the number of datasets being linked increases, so can the number of applicable governance rules and the complexity of governance. Each additional dataset’s unique set of rules increases the volume of

⁴ For example, under HIPAA, PHI retains its privacy and security requirements even when combined with other datasets. If PHI is linked with non-healthcare data, the resulting dataset must still comply with HIPAA’s privacy and security rules. Similarly, a DUA for healthcare data may require that the recipient adhere to HIPAA or other privacy regulations, even if the data is combined with unrelated datasets.

rules to sort through and the likelihood of conflicts and overlaps.

Who Conducts Rule Harmonization?

The coordinating body leads the rule harmonization process with participation from the dataset holder(s)⁵ and the investigator/requesting institution.⁶ When linking multiple datasets, neither the investigator/requesting institution nor the dataset holder(s) are well positioned to lead rule harmonization. Dataset holders are bound by their own policies, which may limit their ability to objectively reconcile rules across multiple datasets. Similarly, while the investigator/requesting institution is central in the research process, their vested interest in the outcomes of the linkage and the study could introduce bias into the harmonization process. Additionally, dataset holders and the investigator/requesting institution often face practical constraints, such as limited time and resources to identify rules from other dataset holders, reconcile conflicts between governance rules, and convene discussions to negotiate rules.

The coordinating body is a neutral entity that is best suited to lead rule harmonization. Operating independently of both the dataset holder(s) and the investigator/requesting institution, the coordinating body promotes impartiality and avoids concerns of bias. Its responsibilities include facilitating collaboration, mediating conflicts, and understanding the linkage governance landscape. To fulfill these duties, the coordinating body must possess expertise in research processes, ethics, data governance, health policy, and legal interpretation.⁷

While the coordinating body leads the effort, rule harmonization is necessarily collaborative. Active engagement throughout this process and input from the dataset holders is essential to address concerns, refine rules, and build trust and accountability among all parties.

Common Governance Process Activities that Precede Rule Harmonization

The Common Governance Process has three stages: Intake, Screening, and Rule Harmonization. Several activities establish the optimal conditions for an efficient Common Governance Process. The investigator/requesting institution must have designed the study, secured approvals from oversight entities like IRB and the Privacy Board, and submitted a complete request package that consolidates study information and initiates the Common Governance Process ([Section 4.1 of Framework Playbook](#)). The request package submission creates the record and starts the workflow within which rule harmonization will occur. The request package identifies requested datasets to be linked, and the governance requirements already reviewed by the investigator/requesting institution. By identifying datasets to be linked, the request package also identifies the dataset holders that will participate in rule harmonization

⁵ Linking multiple datasets means involving multiple dataset holders in a governance process.

⁶ See [Section 5 of Framework Playbook](#) for descriptions of these roles.

⁷ The rule harmonization process recognizes widespread interest in how artificial intelligence can streamline governance by performing key governance tasks. For a number of reasons, rule harmonization as described in this document cannot be performed by a large language model (LLM) at this time. LLMs lack the nuanced understanding required to interpret complex legal, ethical, and institutional requirements. Governance rules often involve subjective judgment, contextual considerations, and negotiation among stakeholders, elements that cannot be fully captured or resolved by an automated system. For example, reconciling conflicts between rules may require weighing competing priorities, understanding the intent behind specific policies, or addressing ethical concerns that demand human empathy and discretion. These tasks are inherently human and require collaboration, critical thinking, and stakeholder engagement.

and the datasets for which sources of governance rules will be gathered.

Because rule harmonization is a resource-intensive process, the dataset holders conduct a preliminary "go/no-go" review to assess whether the study is feasible and permissible (Screening stage, [Section 4.2 of Framework Playbook](#)) before proceeding to the Rule Harmonization stage ([below](#)). Requests that receive a "go" decision from all dataset holders during screening advance to rule harmonization. Those that are "no-go" will not proceed and will receive feedback through the coordinating body.

Rule Harmonization Steps

This section describes the steps for conducting rule harmonization:

1. Gather governance sources that apply to one or more requested datasets or dataset holders.
1. Review each governance source and extract all rules, assembling and organizing into an inventory of governance rules.
2. Classify each inventoried rule by domain and subdomain and determine whether each inventoried rule applies to the creation or use of the linked dataset for research as described in the request package.
3. Compare applicable governance rules across datasets and categorize rule relationship.
4. Propose harmonized rules for overlapping or conflicting rules and compile into a draft consolidated rule set.
5. Circulate draft consolidated rule set for review and resolve issues of concern through collaboration and cooperation.
6. Issue a final approval or denial of the request.

Refer to the rule harmonization overview (Figure 1) for a high-level depiction of the process. The rule harmonization business process diagrams depicting the activities within each of these steps are presented in Figure 2, Figure 3, and Figure 4.⁸ The sections that follow describe each step in detail. Within the Rule Harmonization stage, the dataset holders and the investigator/requesting institution make four key governance decisions (decisions 2-5 in the Common Governance Process; see [Section 6 of Framework Playbook](#)):

- Governance Decision 2: Is the draft rule inventory complete and accurate? (dataset holder)
- Governance Decision 3: Is the draft consolidated rule set acceptable? (dataset holder)
- Governance Decision 4: Can the study comply with the consolidated rule set? (investigator/requesting institution)
- Governance Decision 5: Is the request approved? (dataset holder)

Each decision point is presented in the business process diagrams with a yellow diamond and

⁸ Explanations for Intake and Screening, as well as their business process diagrams, are found in [Section 4 of Framework Playbook](#).

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described in detail below. White diamonds reflect operational decisions that support navigation within the rule harmonization process and are represented distinct from governance decisions because they lack governance implications for the request.

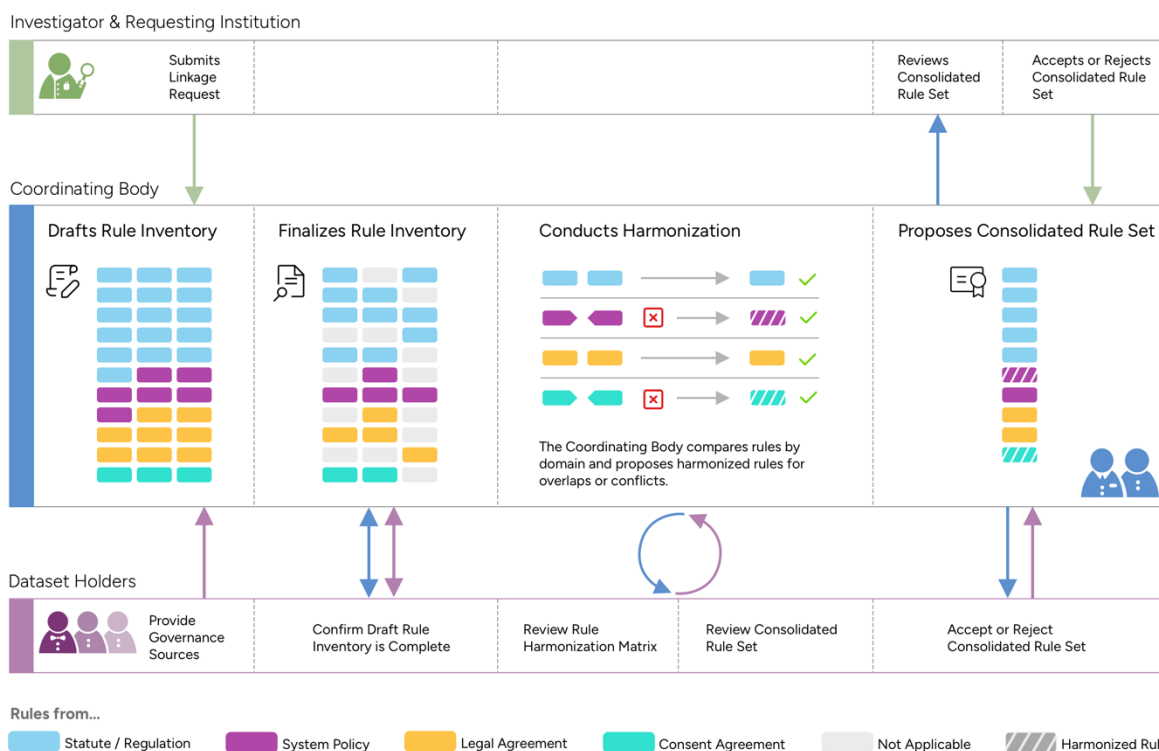


Figure 1: Rule Harmonization Overview

Step 1. Identify and Review Governance Sources

The first step in rule harmonization (see Figure 2) is for the coordinating body to gather the relevant governance sources that apply to the requested datasets identified within the request. Governance sources may be dataset-specific sources or generally applicable governance sources. Examples of dataset-specific sources include informed consents and DUAs. Generally applicable sources are cross cutting such as federal laws governing health data or security that apply across datasets.⁹ Two common examples are the Privacy Act of 1974 and HIPAA.

Relevant refers to a source that contains governance content about research, data use, access, and/or linkage. The coordinating body may have to review a source's content to determine relevance and discard irrelevant sources. Recognizing that governance sources can vary widely in how they are created, structured, and published, a dataset may have just a few or many governance sources, each containing one or more governance rules. Because governance sources often reference other sources, such as a DUA requiring users of a dataset adhere to HIPAA, the coordinating body also retrieves and reviews sources referenced in other

⁹ Some governance sources may not be applicable to research linkage; for example, policies on disaster recovery or system backups are unlikely to be relevant to an investigator requesting access to a dataset.

governance sources. Such referenced sources are often generally applicable governance sources because they offer translatable or common guidance.

The coordinating body catalogs each governance source by recording information about its identity and provenance (e.g., name, version number, date, and information about where it was located). Generally applicable governance sources are cataloged in the same manner as dataset-specific sources. Recognizing that governance sources evolve over time, recording provenance information provides important context on the state of governance at the time of rule harmonization.

Step 2. Inventory and Validate Governance Rules

Once governance sources have been collected, the coordinating body carefully extracts and documents the governance rules each source contains in a rule inventory. The coordinating body enters rules into one rule inventory for the request in a standard format using a structured worksheet (see [Template 1](#)) that retains each rule's originating source in the *Governance Source*, *Source Link*, and *Source Details* columns. To support completeness, traceability, and provenance, the coordinating body develops one rule inventory for the request. Every rule is labeled based on which dataset the rule applies to in the *Dataset* column. Rules that apply to multiple datasets are labeled as **generally applicable governance rules**.

The coordinating body then assesses whether each rule applies to the creation or use of linked data (i.e., applies to the request) and assigns the rule one of three values: "applicable," "not applicable," or "unable to determine." In the notes field, the coordinating body records a brief explanation of how the rule does or does not apply to the request. Notably, rule applicability may or may not be apparent or easily discernable to the coordinating body.

Rules assessed by the coordinating body as inapplicable to the request remain documented in the rule inventory for completeness, provenance, and traceability, but will be excluded from subsequent harmonization steps. By retaining inapplicable rules in the rule inventory, dataset holders who review the rule inventory are able to verify that the entire governance source was reviewed and all rules were extracted, confirm that inapplicable rules are accurately excluded, assess applicability for rules where the coordinating body was unable to determine, and correct erroneously excluded rules. Applicability is a critical attribute because it determines which rules will be included in the harmonization and ultimate consolidated rule set.

The coordinating body assigns every rule in the inventory to a domain and subdomain (*Rule Domain* and *Rule Subdomain* columns). This protocol recommends using each action in the data lifecycle as a domain, and relevant subtopics within that action as subdomains; for example, data access as a domain and access controls as one subdomain of data access (see [Template 2](#)). Every domain and subdomain may have zero, one, two, or many rules and these patterns will vary by dataset. Domains and subdomains are a tool to organize rules the same way across datasets, so that rules with similar application can be grouped and compared across datasets in the next step. This step marks the completion of the draft rule inventory.

The rule inventory is considered a draft until reviewed and accepted by the dataset holders. The process is now at Governance Decision #2: Is the draft rule inventory complete and accurate? The coordinating body shares the draft rule inventory with each dataset holder to confirm that

the governance sources are current, the inventory is complete, the extracted rules and provenance information are accurate, the applicability determinations are accurate, and the domain and subdomain assignments are appropriate. Dataset holders may also comment on generally applicable rules as needed.

Each dataset holder either accepts or does not accept the draft rule inventory as accurate and complete. When not accepted, the dataset holder provides revisions on the draft rule inventory to the coordinating body. Once the draft rule inventory is accepted by the dataset holders, the coordinating body updates the rule inventory as final with the understanding that dataset holders may provide further feedback and changes as needed.

Step 3. Map and Categorize Rules for Harmonization

Once the rule inventory is finalized, the coordinating body compares and maps rules across datasets by domain and subdomain using a rule harmonization matrix (see [Template 3](#)). Mapping begins with an applicability filter. Only rules from the final inventory marked as applicable are mapped.

The coordinating body uses the domain and subdomain assignments to create a group of rules for comparison and mapping. Because a dataset may have zero, one, or multiple rules in a domain/subdomain, the mapping exercise may be a comparison of two, three, or more rules. The coordinating body maps the groups of governance rules based on relationships between rules as follows:

- **Duplicate Rules:** Rules that are identical across datasets. These rules are deduplicated and one flows through unchanged to the consolidated rule set.
- **Unique Rules:** Rules that apply to only one dataset and do not align with rules from other datasets. These rules flow through unchanged to the consolidated rule set.
- **Overlapping Rules:** Rules that address the same or a closely related objective and impose similar requirements or achieve the same outcome, but are not identical in wording, specificity, or application. Overlapping rules interact in practice – by reinforcing, conditioning, narrowing, or supplementing one another – such that compliance requires an integrated reading rather than treating one rule as a complete substitute for the other. Overlap includes instances where one rule changes how another operates, and instances where the rules together create additive procedural, documentation, technical, or oversight requirements. Overlapping rules require harmonization to ensure clarity on investigator/requesting institution obligations while eliminating – where possible – duplicative and unclear requirements.
- **Conflicting Rules:** Rules that impose contradictory requirements that cannot be simultaneously satisfied. For example, one dataset may require data to be encrypted

Example of Overlapping Rules: One dataset may require that data be stored in "secure, access-controlled environments," while another specifies that data must be stored in "restricted systems with role-based **access controls**." Although phrased differently, both rules describe the same underlying access/security requirement.

using a specific algorithm, while another may require a different encryption standard for compliance with its governance policies. Conflicting rules require harmonization to find a middle ground, or they may be determined to be impossible to link.

Mapping results in a matrix of rules organized by domain/subdomain where each grouping of rules is assigned to one or multiple categories based on how they interrelate. The assigned categorization directs the coordinating body on what to do next. Duplicate and unique rules will simply flow into the consolidated rule set. Overlapping and conflicting rules will be harmonized in the next step.

Generally applicable rules are mapped alongside dataset-specific rules when they address the same topic. If a generally applicable rule is duplicative with dataset-specific rules, it may be deduplicated. If it establishes a stricter baseline requirement, it may shape the harmonized rule. It is possible that the rule inventory will surface internal rule conflicts, where multiple rules for the same dataset from different governance sources are in conflict with each other.

Detailed documentation of rule mapping is critical for maintaining transparency and provenance throughout the harmonization process. Dataset holders may reference mapping documentation when evaluating how their rules were categorized and harmonized with others. This documentation will serve as a valuable reference for replicability, updates, or modifications to the governance framework, ensuring that future harmonization efforts can build on prior decisions and insights.

Step 4. Draft Harmonized Rules and Consolidated Rule Set

For overlapping and conflicting rules, the coordinating body drafts harmonized rules for consideration by the dataset holders:

- For overlapping rules, the coordinating body may select one of the overlapping rules that may be acceptable to all dataset holders or draft a new harmonized rule that serves as a middle ground, blending aspects of the rules that overlap.
- For conflicting rules, the coordinating body drafts a single harmonized rule that may reflect the shared intent of the conflicting rules and aims to be acceptable by dataset holders. Conflicting rules may require the coordinating body to consult the policies and governance sources to consider how much flexibility is available.

Harmonized rules should balance the needs of all dataset holders and be grounded in the framework's guiding principles and standards. When possible, the coordinating body will reference rule harmonization materials from past successful linkages, where similar rule conflicts were overcome. The coordinating body then drafts a consolidated rule set by integrating harmonized, duplicate, and unique rules organized by actions in the data lifecycle (e.g., rules about data access).

Step 5. Engage Dataset Holders about Acceptance of the Consolidated Rule Set

The coordinating body shares with the dataset holders the draft rule harmonization matrix ([Template 3](#)) and the draft consolidated rule set. The rule harmonization matrix intentionally preserves the provenance of rules and sources throughout the rule harmonization process so that the dataset holder can trace how the consolidated rule set was created. Sharing all

materials across datasets should support collaboration across dataset holders and drive agreement on harmonized rules that will allow data linkage to occur. The process is now at Governance Decision #3: Is the draft consolidated rule set acceptable?

The precise criteria for acceptability will vary between dataset holders but generally means containing rules that are clear, enforceable, compliant with applicable policies, and ready for implementation. Each dataset holder reviews the draft consolidated rule set and either accepts or does not accept it. As part of review, other individuals within the dataset holder's organization such as technical experts may evaluate compatibility with the systems and processes used to manage the linked dataset, while legal counsel may be needed to evaluate the harmonized rules for compliance with applicable laws. Once the dataset holders agree that the rules are clear, enforceable, and ready for implementation, they accept the draft consolidated rule set. By accepting, the dataset holder is also affirming their willingness to sign a DUA with these rules included as terms. Once the coordinating body has received acceptances from all dataset holders, the consolidated rule set is no longer a draft, but it is not yet finalized.

In cases where harmonized rules are not acceptable, the coordinating body should facilitate exchanges between dataset holders to seek solutions, modifying harmonized rules based on feedback from each dataset holder until the consolidated rule set is accepted by all dataset holders, or a determination is made that it is not feasible to achieve a consolidated rule set. Discussions between dataset holders are facilitated by the coordinating body, which is responsible for facilitating this collaborative and iterative process, ensuring that discrepancies or conflicts are resolved. Dataset holder cooperation and collaboration may occur in asynchronous communication or as a group.

In cases where the consolidated rule set cannot be accepted despite efforts between the dataset holders, the coordinating body will prompt the dataset holders to deny the request.

Step 6. Confirm Study Compliance with Consolidated Rule Set

For a consolidated rule set that has been accepted by all dataset holders, the coordinating body sends the consolidated rule set to the investigator/requesting institution to confirm the study's ability to comply (see Figure 4). The process is now at Governance Decision #4: Can the study comply with the consolidated rule set? If the investigator/requesting institution can comply, they send an acceptance to the coordinating body. During this time, the coordinating body is responsible for responding to any questions with clarifications.

If compliance is not possible, the investigator/requesting institution identifies which rules they cannot comply with and determines whether or not they will alter the study (e.g., remove a dataset), seek a change in the consolidated rule set, or cancel the request (an operational decision in the process). The investigator/requesting institution then sends the feedback and decision to the coordinating body. Determining whether or not to cancel may depend on the type of rule conflicts in the consolidated rule set.

When rule feedback is received without a decision to cancel the request, the coordinating body revisits the consolidated rule set with revisions. When rule feedback is received with a decision to cancel, the coordinating body updates the request status to canceled and notifies the dataset holders.

Step 7. Decide on the Request for Data Linkage

The dataset holders make the final governance decision to approve or deny the request to link their dataset(s) during the last step in rule harmonization (see Figure 4). The coordinating body updates the request status to ready for decision and that action notifies the dataset holders, prompting them to conduct their final request review. The process is now at Governance Decision #5: Is the request approved? This decision is the formal approval by the dataset holders to permit access to and use of their dataset for the proposed linkage under the terms set forth in the consolidated rule set.

Approve the Request: The dataset holders conduct their final review of the request package and consolidated rule set, before approving the request to proceed. For requests that have been approved in screening and have an accepted consolidated rule set, approval by dataset holders is expected.

The dataset holders send a formal approval to the coordinating body. Once approved by all dataset holders, the coordinating body marks the request as approved and notifies the investigator/requesting institution. This marks the end of the rule harmonization process for an approved request. The dataset holders may proceed with creating a DUA(s) based on the rules established in the consolidated rule set to start the study.

Deny the Request: If the consolidated rule set could not be accepted by all dataset holders and a resolution to eliminate rule conflicts cannot be reached, the dataset holders deny the request and share the rationale for denial with the coordinating body as request feedback. On receipt of a denial, the coordinating body marks the request as denied and communicates that decision and feedback to the investigator/requesting institution.

The investigator/requesting institution receives and reviews feedback and then determines if they can and will revise the study or the linkage implementation in order to address the rule conflict and resubmit the request package. If revision and resubmission is pursued, the investigator/requesting institution returns to the Intake stage. This marks the end of the rule harmonization process for a denied request.

A denied request should be treated as an important learning opportunity for the Linkage Governance Framework. These insights can help inform future governance processes and improve the feasibility of linking datasets for research. When this occurs, the coordinating body should capture a description of why data linkage could not be accomplished and, where possible, disseminate recommendations to encourage change in dataset governance that could facilitate future linkage.

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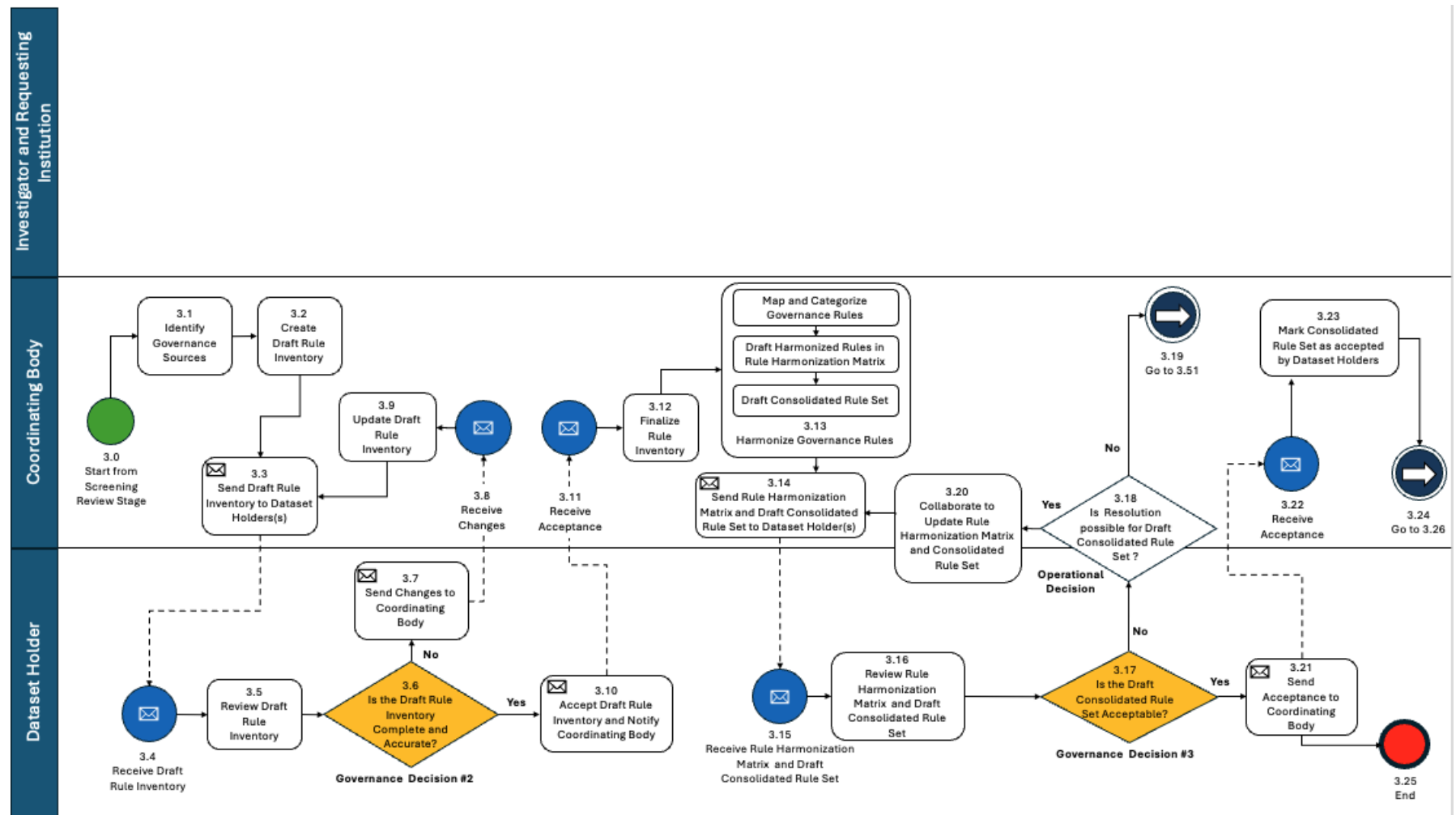


Figure 2: Rule Harmonization Business Process Diagram (1 of 3)

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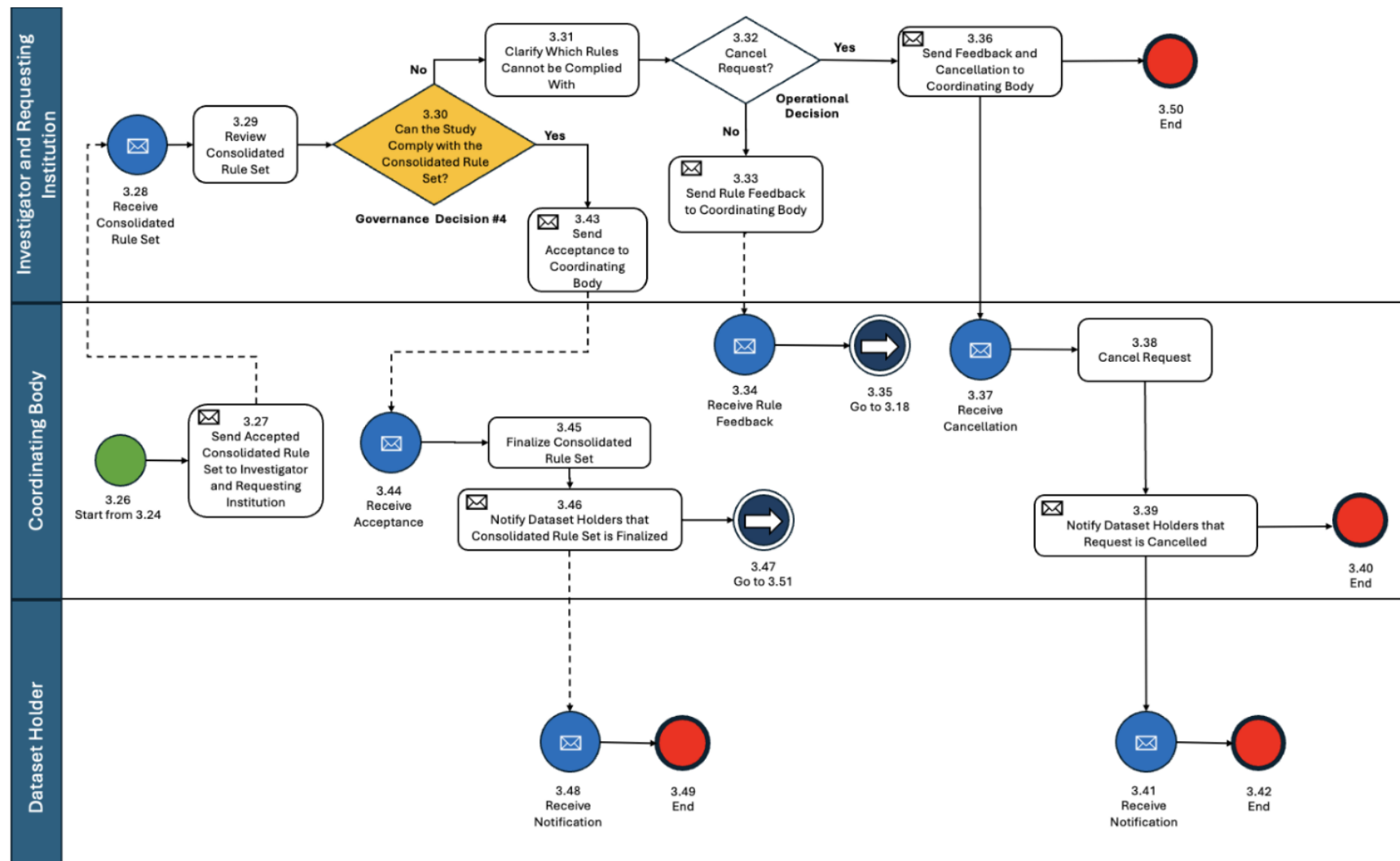


Figure 3: Rule Harmonization Business Process Diagram (2 of 3)

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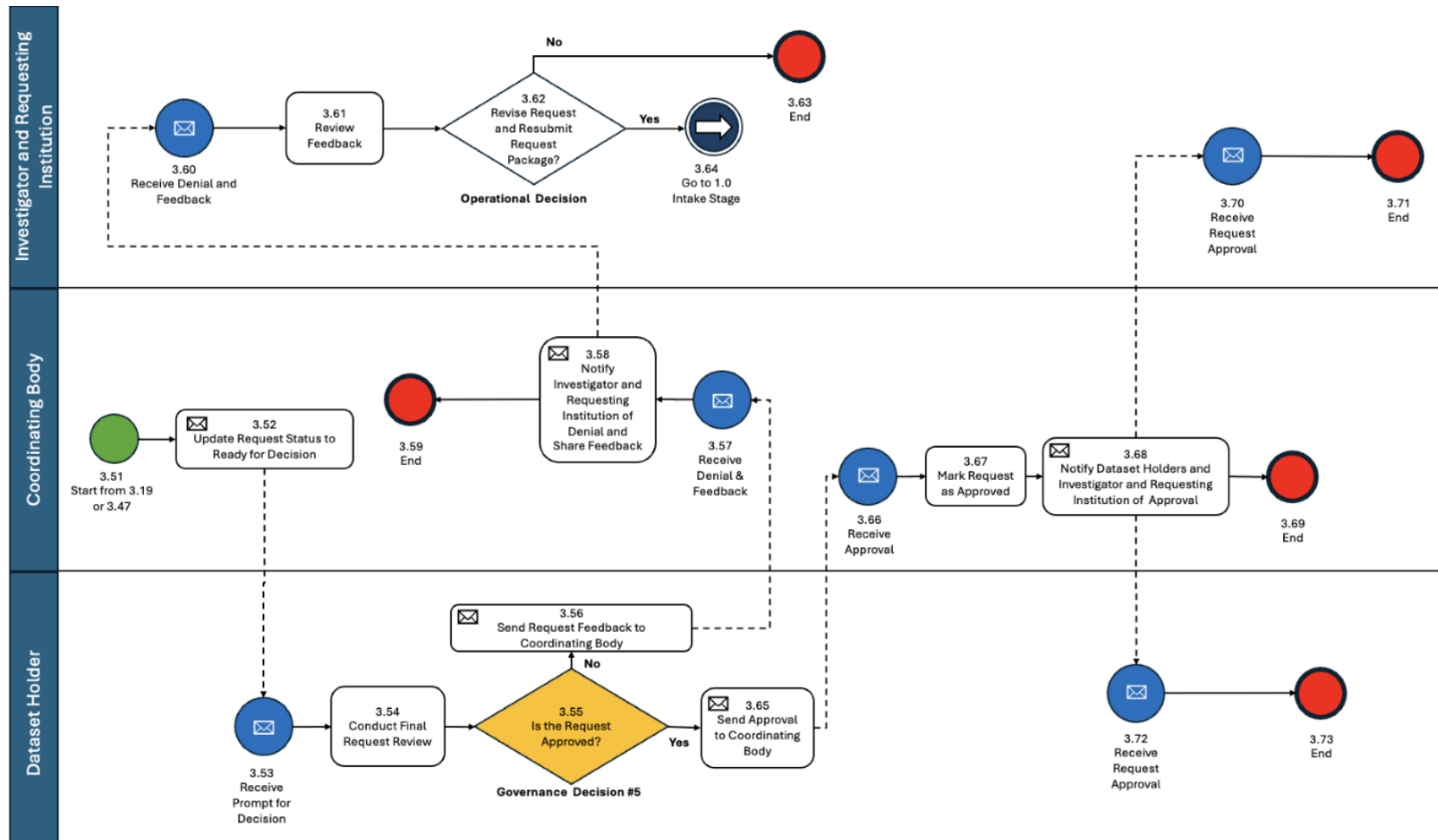


Figure 4: Rule Harmonization Business Process Diagram (3 of 3)

Foundation of Governance Rules

Governance rules are grounded in their origin, intent, and/or application. The origin refers to the law, policy, regulation, etc. where the rule comes from. The intent refers to the purpose of the rule, and the application refers to who and how the rule accomplishes its desired intent.

Origin of Governance Rule

Governance rules can originate from a variety of types of sources including but not limited to federal, state, local, or other laws, regulations, or policies; institutional policies; project, study, or dataset policies; ethical requirements; and operational processes or norms. Some governance sources and rules are specific to an individual dataset, while others are generally applicable across datasets when they govern the linked dataset context. Generally applicable governance sources may include cross-cutting laws, standards, or policies that apply to multiple datasets.

Laws: Laws define legal requirements that are among the most critical considerations in rule harmonization, as they impose a variety of obligations at many different levels (e.g., obligation for an institution, a person, or an information system). Laws such as HIPAA govern the handling of PHI and may dictate strict requirements surrounding de-identification, access controls, or security measures that must be met to maintain compliance. While laws define what the requirements are, they may not necessarily dictate how the requirements are to be met. Laws also specify when and to what the law does and does not apply and exceptions to the law. In this example, HIPAA is the origin of many policies and is organized into sections that create hundreds of rules applicable to different parts of the data lifecycle. For example, a hospital's data may need to adhere to HIPAA's Privacy Rule and Security Rule, which mandate safeguards to protect sensitive information.

Laws may overlap with other laws, where one law references another or one builds on the obligations set by another. An example is state-specific laws, such as the California Consumer Privacy Act, that impose additional restrictions beyond HIPAA, such as stricter consent requirements or limitations on data sharing.

Legal requirements are non-negotiable and must be strictly adhered to during rule harmonization. Failure to comply with legal requirements can result in severe penalties, including fines, sanctions, and reputational damage.

Institutional Policies: Institutional policies are the policies set by an organization that reflect its priorities, risk tolerance, and operational practices. These policies may include procedural requirements, such as requiring a data governance council to review and approve all external data-sharing activities. For example, one institution may require data sharing to occur via its own secure file transport protocol and define the conditions that permit remote access. Organizations may have one comprehensive policy document or many more targeted policy documents. Institutional policies often build on or reference laws such as HIPAA or standards such as National Institute of Standards and Technology Special Publication 800-53 that defines security controls for federal information systems. For example, while all hospitals may have policies requiring compliance with HIPAA, a hospital may specify which de-identification methods permitted under HIPAA are acceptable for its datasets.

Project-, Study-, or Dataset-Specific Policies: Projects, studies, or datasets may establish unique policies that govern the data associated with a study, project, or dataset. These policies may build on institutional policies at the organization where the project study or dataset resides and may address a variety of issues specific to the project, study, or dataset such as a study's consent requirements, data usage restrictions on specific data assets, or the criteria that require additional engagement with the IRB. For example, a dataset's use may be restricted to specific topics or domains of research, prohibit use by for-profit companies, or require a specific attribution in all resulting publications.

IRB or Privacy Board Directives or Policies: Certain governance rules originate from an IRB or Privacy Board's directives or policies, informed consent documents, or processes. For an IRB, an example would be the process for notifying the IRB about an unanticipated deviation from the study protocol. Regarding informed consents, datasets collected as part of a research study will have a dedicated consent document that stipulates rules about how the research data can be used, whereas data from routine healthcare delivery usually falls under a broader institutional consent that covers all data practices. Example consent rules from research-created data are that a person's data cannot be linked without reconsenting them for linkage on a study-by-study basis.

Operational Practices or Norms: Operational practices or norms are the informal, procedural, or culturally ingrained methods organizations use to manage data. These practices are not always formally documented but influence how governance rules are implemented throughout the data lifecycle. For example, one dataset holder may back up all their data nightly. When datasets in their environment need to be destroyed, backup copies of the dataset may also need to be considered as part of the destruction process.

Intent of Governance Rule

Every governance rule is designed to achieve one or multiple objectives, which reflect the underlying priorities of the originating policies. Intentions for governance rules may include transparency, data minimization, purpose specification and use limitation, privacy, security, consistency and standardization, accountability, and replicability.

Transparency: Transparency means that the organization is open and clear about how the data is accessed, shared, linked, used, or disseminated. This includes providing clear information about consent, data provenance, and usage restrictions. Transparency empowers data users by informing them of the purposes of data collection and their rights regarding its use. It also supports compliance with legal and ethical standards that require explicit disclosure of data practices.

Transparency also promotes findability and accessibility. For example, metadata that documents consent agreements, data origins, and usage restrictions makes the data easier to locate and understand, while ensuring that stakeholders are informed about its governance.

Data Minimization: Data minimization is a key directive in HIPAA¹⁰ and focuses on collecting

¹⁰ HIPAA Minimum Necessary Requirement: <https://www.hhs.gov/hipaa/for-professionals/privacy/guidance/minimum-necessary-requirement/index.html>.

and using only the data necessary to achieve the intended purpose of the linkage. By limiting the scope of the data, this reduces the risk of privacy breaches, minimizes the potential for misuse, and ensures compliance with regulations that require proportionality in data collection and use. Data minimization also supports efficient data management, as smaller datasets are easier to secure and analyze. Data minimization complements interoperability, as smaller, well-defined datasets are easier to integrate and analyze across systems while maintaining ethical and legal constraints.

Purpose Specification and Use Limitation: Purpose specification and use limitation means clearly defining the reasons for collecting data and restricting its use to those stated purposes unless additional consent is obtained. This principle prevents unauthorized or unethical uses of data, aligns linkage with original objectives, and complies with applicable regulations. By adhering to defined purposes, organizations safeguard the integrity of the data and reduce risks associated with misuse.

Privacy: Privacy refers to the right of individuals to control how their personal health data is collected, used, shared, and stored. It ensures that sensitive health details, such as diagnoses, treatments, and medical history, are protected from unauthorized access or disclosure. Health data privacy involves safeguarding confidentiality while respecting patients' autonomy and their ability to make informed decisions about who can access their information. Privacy is also about ensuring individuals feel safe sharing details about their medical history and conditions without fear of judgment or misuse. Safeguarding health data privacy helps prevent unauthorized access, reducing the risk of identity theft, discrimination, or exploitation based on medical information. Examples of privacy rules include data anonymization and small cell suppression. For example, privacy-enhancing techniques, such as de-identification or encryption, must be applied consistently across the linked dataset to maintain compliance with applicable regulations and ethical standards.

Security: Security involves implementing measures, policies, and technologies designed to protect data from unauthorized access, breaches, loss, or misuse. It safeguards the confidentiality, integrity, and availability of data throughout its lifecycle, using security measures such as encryption, access controls, and regular audits. Security also enhances accessibility, by implementing mechanisms so that only authorized users can access data safely while preventing unauthorized access or breaches. Security rules focus on safeguarding data from unauthorized access, breaches, or misuse. These rules may include requirements for encryption, access controls, regular audits, or secure storage environments. For example, one dataset may require data to be stored in encrypted formats, while another may mandate additional physical security measures for servers.

Consistency and Standardization: Consistency and standardization refer to the alignment of governance rules and technical requirements to ensure seamless data integration, sharing, and management across datasets. This involves harmonizing elements such as data formats, encryption protocols, and interoperability standards to create a unified framework that minimizes errors and inconsistencies. For example, one dataset may store information in a particular file format, while another uses a widely accepted open standard.

Accountability: Accountability refers to the obligation of investigators, organizations, and systems to take responsibility for their actions, decisions, and compliance with established rules

and standards. It involves defining roles, responsibilities, and enforcement procedures along with creating mechanisms to monitor, enforce, and report adherence to governance frameworks. By holding all parties answerable for their roles in managing and protecting data, accountability fosters a culture of transparency, ethical behavior, and responsibility in data-sharing initiatives. Accountability also supports findability, as clear documentation of roles and responsibilities helps stakeholders identify who is responsible for specific aspects of data governance and management.

Replicability: Replicability refers to the ability to consistently replicate a research study or governance process and reproduce research findings by ensuring that data management practices, processes, and documentation are standardized, transparent, and well-maintained. It involves establishing clear protocols for data collection, storage, sharing, and analysis so that other researchers can access and use the same data in a reliable and repeatable manner. Rules about replicability ensure that governance processes are open and well-documented, allowing others to understand how data was handled and analyzed. It supports verification of the integrity of research findings by enabling others to reproduce results using the same data and governance practices. Replicable governance supports seamless sharing and re-use of data across research teams, fostering collaboration and accelerating scientific progress.

Application of Governance Rule

Governance rules can apply to different parties (persons or organizations) and to different actions that party may take. In linking data for research, governance rules can apply to one or more steps in the data lifecycle: access, sharing, linkage, use, and dissemination.

Data Access: Data access is acquiring data from a data repository or other system to store, manage, and share data for reuse. Authorized persons or entities may gain entry to datasets under governance rules that outline eligibility criteria, approval processes, and monitoring requirements. These rules define who can access the data, under what conditions, and through what mechanisms, ensuring compliance with security and governance requirements while protecting sensitive information. For example, one dataset may require access to occur only within secure environments, while another may permit remote access with multi-factor authentication or transmission of data to an investigator on physical media.

Data Sharing: Data sharing refers to making data available to the broader data user community, such as submitting data to a repository for dissemination or sending data to an investigator for an approved research study. It involves granting access to data between entities under governance rules that specify recipients, conditions, mechanisms, and restrictions. These rules define who data can be shared with, the conditions for sharing, the parameters for safe and secure transmission, and the safeguards required to ensure compliance with privacy, ethical, and legal standards. For example, one dataset may limit sharing to academic institutions, while another may permit sharing with government agencies under specific agreements.

Data Linkage: Data linkage refers to the process of combining data from two or more datasets at the individual level to create a unified dataset. Rules governing data linkage may define if data can be linked, processes for requesting linkage, what linkage methods are permissible, required technical standards, and what parameters are required when conducting linkage. For

example, a hospital may allow linkage only using Privacy-Preserving Record Linkage so that no identifiers are shared and prohibit re-identification of the individuals within the linked dataset.

Data Use: Data use refers to working with data for secondary research or other analytical purposes. Rules on data use can specify how linked data can be used, permissible research purposes, and restrictions on analysis. For instance, one dataset may limit use to disease-specific research, while another may allow broader applications. Since data use reflects a sequence of tasks including data management, storage, and retention, data use rules can dictate how long data can be kept and under what conditions it must be deleted. For example, one dataset may require retention for the period of the study, while another mandates deletion after a specific timeframe.

Data Dissemination: Data dissemination refers to the process of distributing data used in or created from research in its original or derivative form to the public, including other investigators, clinicians, and other partners, and is distinct from dissemination of research findings. In the case of linkage, dissemination refers to distribution of the originating datasets and linked dataset. Rules govern how data can be additionally distributed to authorized parties or the public, including the conditions, methods, and safeguards required to ensure compliance with privacy, ethical, and legal standards. For example, some linked datasets will be prohibited from further distribution while others will be created with extensive reuse as an explicit goal such that intentional rules and processes will dictate how the linked dataset can be disseminated for widespread use.

Template 1. Rule Inventory

The rule inventory provides a format to create a detailed record of governance rules, organized by domain and subdomain. This template (see Table 1) ensures that all rules are documented systematically, with clear references to their governance sources and provenance. The Coordinating Body produces one rule inventory for the request containing rules from all requested datasets and any generally applicable governance sources relevant to the linked dataset context. In addition to domain and subdomain classification, each rule is assigned an applicability designation.

Table 1: Rule Inventory Template

Rule ID	Applicability (Y/N)	Rule Domain	Rule Subdomain	Dataset	Rule	Governance Source	Source Link	Source Details	Notes
A-14	Y	Data Access	Access Controls	[insert dataset name]	Data must be stored in secure environments.	Institutional Policy	Link; 06-15-2025	Policy Version 2.0, Section 4.3, Updated 2024	Aligns with HIPAA Security Rule; applies because the linked dataset will be stored by the requesting institution
A-22	N	Data Sharing	Recipients	[insert dataset name]	Data may be shared only with government agencies.	DUA	Link; 06-15-2025	DUA Version 1.1, Section 2.1, Signed 2024	Does not apply because the proposed linked dataset will not be shared onward to government agencies; retained for provenance.
GA-10	Y	Data Use	Permissible Uses	Generally Applicable	Data may be used only for public health research.	Federal Policy	Link; 06-15-2025	Policy Version 1.0, Section 3.2	Applies because the study purpose falls within the linked dataset use context.

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Rule Harmonization Protocol

Complete the columns in the rule inventory template as follows:

- **Rule ID:** A unique identifier assigned to each inventoried rule to support traceability across the rule inventory, rule harmonization matrix, and consolidated rule set.
- **Applicability (Y/N):** Indicates whether the rule applies to the linked dataset context for the request.
- **Rule Domain:** The broad governance area the rule falls under (e.g., Data Access, Data Sharing, Data Linkage, Data Use, Data Dissemination).
- **Rule Subdomain:** A more specific topic within the domain (e.g., Access Controls, Recipients, Permissible Purposes).
- **Dataset:** The dataset to which the rule applies. This field should also support the value "Generally Applicable" for cross-cutting rules that govern the linked dataset context regardless of dataset origin.
- **Rule:** The specific directive or guideline governing the dataset (e.g., "Data must be stored in secure environments"). The rule should be written verbatim or paraphrased from the governance source for accuracy.
- **Governance Source:** The source from which the rule originates (e.g., federal law, institutional policy, DUA, consent form).
- **Source Link:** Hyperlink for where the source was retrieved from and accessed date.
- **Source Details:** Additional information about the governance source, such as version/date, relevant sections, and any references to external policies.
- **Notes:** Contextual information or observations about the rule (e.g., "Aligns with HIPAA Security Rule," "Stricter sharing than Dataset A"). Should include the trigger condition used to determine applicability and a brief explanation of how the rule does or does not apply to the linked dataset context.

Template 2. Proposed Rule Harmonization Domains and Subdomains

The rule harmonization domains and subdomains provide a proposed method for organizing governance rules across datasets. Table 2 organizes the rules into key areas of the data lifecycle: data access, sharing, use, linkage, dissemination, retention, and technical standards. Each domain is further refined into suggested subdomains to group rules by specific topics (e.g., access controls, permissible research purposes). These subdomains are not exhaustive and serve as a starting point for structuring governance rules. During the rule harmonization process, additional subdomains may emerge based on the unique characteristics of the datasets, governance sources, and specific requirements of the study.

Table 2: Proposed Rule Harmonization Domains and Subdomains

Domain	Subdomains
Data Access	Access Controls User Eligibility Access Mechanisms Access Approval Process Access Monitoring and Auditing Access Duration
Data Sharing	Recipients Conditions Mechanisms Restrictions Accountability/Oversight
Data Use	Permissible Uses Prohibited Uses Analysis Restrictions Dissemination of Results Ethical Oversight Data Use Transparency
Data Linkage	Methods Approval Feasibility Safeguards Accountability/Oversight Legal and Ethical Considerations

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Domain	Subdomains
Data Dissemination	Recipients Conditions Mechanisms Restrictions Accountability/Oversight Dissemination Transparency
Data Retention	Duration Conditions Safeguards Accountability
Technical Standards	Data Formats Encryption Protocols Interoperability Requirements System Security Data Quality Data Validation

Template 3. Rule Harmonization Matrix

The rule harmonization matrix is a tool to document the process of mapping governance rules across datasets, categorization of relationships between rules, and proposed harmonized rules. The rule harmonization matrix captures rules from originating datasets, the category that best describes how rules are related and rationale, and the proposed harmonized rule and rationale. Only rules marked Applies = Y in the rule inventory are included into the rule harmonization matrix. The rule harmonization matrix should preserve traceability to the rule inventory by including each rule's Rule ID within the applicable dataset rule field(s). Table 3 illustrates how the rule harmonization matrix can be completed with four examples.

Table 3: Rule Harmonization Matrix

Rule Domain – Subdomain	Dataset A Rule(s)	Dataset B Rule(s)	Dataset C Rule(s)	Generally Applicable Rule(s)	Mapping Category	Mapping Categorization Rationale	Proposed Harmonized Rule	Rationale for Harmonization
Data Access – Access Controls	A-14: Data must be stored in secure environments with restricted access.	B-9: Data must be stored in restricted systems with role-based access controls.	C-6: Data must be stored in secure environments with multi-factor authentication.	GA-03: Storage must meet baseline federal security requirements	Overlap	All rules require secure environments but differ in specifics (e.g., restricted access, role-based permissions, authentication). The Generally Applicable rule establishes a baseline security requirement.	Data must be stored in secure, access-controlled environments that meet restricted access and authentication requirements.	Aligns with stricter security requirements while accommodating all datasets' primary intent.
Data Sharing – Recipients	A-22: Academic institutions only.	B-18: Academic institutions only.	C-11: Academic institutions only.	GA-07: No broader downstream sharing permitted unless expressly authorized.	Duplicate	All datasets specify identical rules allowing sharing only with academic institutions. The Generally Applicable rule reinforces the same limitation.	Allow sharing with academic institutions only.	No harmonization needed; all rules are identical.

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Rule Domain – Subdomain	Dataset A Rule(s)	Dataset B Rule(s)	Dataset C Rule(s)	Generally Applicable Rule(s)	Mapping Category	Mapping Categorization Rationale	Proposed Harmonized Rule	Rationale for Harmonization
Data Use – Permitted Use	A-31: Public health research only.	B-24: Broad use permitted.	C-19: Public health research only.	GA-10: Use limited to approved study purpose.	Conflict	Datasets A and C restrict use to public health research. Dataset B allows broader use. The Generally Applicable rule supports a purpose-limited approach.	Restrict use to public health research.	Aligns with consent form restrictions from Datasets A and C.
Data Use – Retention	A-40: Data must be retained for a minimum of 5 years.	No retention requirements specified.	No retention requirements specified.	No Generally Applicable rule identified.	Unique	Dataset A has a specific retention rule, while Datasets B and C have no requirements.	Retain data for a minimum of 5 years.	Dataset A's rule flows into the consolidated rule set unchanged.

Complete the columns in the rule harmonization matrix as follows:

- **Rule Domain:** The broad governance area the rule falls under (e.g., Data Access, Data Sharing, Data Use).
- **Rule Subdomain:** A more specific topic within the domain (e.g., Access Controls, Recipients, Permissible Purposes).
- **Dataset A Rule(s):** The specific governance rule applicable to Dataset A, as documented in the rule inventory (e.g., Data must be stored in secure environments, or data must be stored in restricted systems with role-based access controls). Include the Rule ID in line with the rule text to preserve traceability to the rule inventory.
- **Dataset B Rule(s):** The specific governance rule applicable to Dataset B, as documented in the rule inventory. Include the Rule ID in line with the rule text to preserve traceability to the rule inventory.
- **Dataset C Rule(s):** The specific governance rule applicable to Dataset C, as documented in the rule inventory. Include the Rule ID in line with the rule text to preserve traceability to the rule inventory.
- **Generally Applicable Rule(s):** The specific applicable governance rule(s) drawn from generally applicable governance sources, when relevant to the same lifecycle area. Include the Rule ID in line with the rule text to preserve traceability to the consolidated rule inventory.

- **Mapping Category:**
 - **Duplicate Rules:** Rules that are identical across datasets. These rules are deduplicated and flow directly into the consolidated rule set without harmonization.
 - **Unique Rules:** Rules that apply to only one dataset and do not overlap with rules from other datasets. These rules flow directly into the consolidated rule set without harmonization.
 - **Overlapping Rules:** Rules that are semantically different but materially the same; they may use different wording, phrasing, or other specifics but impose comparable requirements or achieve the same outcomes.
 - **Conflicting Rules:** Rules that impose contradictory requirements that cannot be simultaneously satisfied.
- **Mapping Categorization Rationale:** A detailed explanation of the relationship between the rules from all datasets and any generally applicable sources, noting whether they overlap or conflict. The rationale should clearly identify the specific differences or similarities between rules.
- **Proposed Harmonized Rule:** The rule proposed to reconcile the overlap or conflict among all the datasets' rules.
- **Rationale for Harmonization:** The justification for the proposed harmonized rule, explaining why it was chosen and how it aligns with governance principles, legal requirements, and ethical standards.